Naïve Bayes: Spam Filter Worked Example

Training Data

We use a toy vocabulary of 5 words: {`free`, `click`, `meeting`, `report`, `offer`}, and $N = 6$ labelled training emails ($x_j = 1$ if word j appears, 0 otherwise). There are $K = 2$ classes: spam ($k = 1$) and ham ($k = 0$).

Email	free	click	meeting	report	offer	Label
1	1	1	0	0	1	**Spam** ($y = 1$)
2	1	0	0	0	1	**Spam** ($y = 1$)
3	0	1	0	0	0	**Spam** ($y = 1$)
4	0	0	1	1	0	**Not spam** ($y = 0$)
5	0	0	1	0	0	**Not spam** ($y = 0$)
6	0	0	0	1	0	**Not spam** ($y = 0$)

There are **3 spam** emails ($y = 1$, i.e. $k = 1$) and **3 legitimate** (ham) emails ($y = 0$, i.e. $k = 0$).

Raw MLE Estimates

The MLE for each parameter is the natural frequency count:

$$\hat{\phi}_{j|k} = \frac{\sum_{i=1}^{N} \mathbf{1}[x_j^{(i)} = 1 \wedge y^{(i)} = k]}{\sum_{i=1}^{N} \mathbf{1}[y^{(i)} = k]}, \qquad \hat{\phi}_y = \frac{\sum_{i=1}^{N} \mathbf{1}[y^{(i)} = 1]}{N}.$$

Prior

$$\hat{\phi}_y = \frac{3}{6} = 0.5$$

Spam class likelihoods ($y = 1$) – denominator: 3

$$\hat{\phi}_{\texttt{free}|1} = \tfrac{2}{3} \approx 0.667 \qquad \text{(emails 1, 2 contain ‘free’)}$$
$$\hat{\phi}_{\texttt{click}|1} = \tfrac{2}{3} \approx 0.667 \qquad \text{(emails 1, 3 contain ‘click’)}$$
$$\hat{\phi}_{\texttt{meeting}|1} = \tfrac{0}{3} = \mathbf{0.000} \qquad \text{‘meeting’ never in spam} \ \bigstar$$
$$\hat{\phi}_{\texttt{report}|1} = \tfrac{0}{3} = \mathbf{0.000} \qquad \text{‘report’ never in spam} \ \bigstar$$
$$\hat{\phi}_{\texttt{offer}|1} = \tfrac{2}{3} \approx 0.667 \qquad \text{(emails 1, 2 contain ‘offer’)}$$

Ham class likelihoods ($y = 0$) – denominator: 3

$$\hat{\phi}_{\texttt{free}|0} = \tfrac{0}{3} = \mathbf{0.000} \qquad \text{‘free’ never in ham} \ \bigstar$$
$$\hat{\phi}_{\texttt{click}|0} = \tfrac{0}{3} = \mathbf{0.000} \qquad \text{‘click’ never in ham} \ \bigstar$$
$$\hat{\phi}_{\texttt{meeting}|0} = \tfrac{2}{3} \approx 0.667 \qquad \text{(emails 4, 5 contain ‘meeting’)}$$
$$\hat{\phi}_{\texttt{report}|0} = \tfrac{2}{3} \approx 0.667 \qquad \text{(emails 4, 6 contain ‘report’)}$$
$$\hat{\phi}_{\texttt{offer}|0} = \tfrac{0}{3} = \mathbf{0.000} \qquad \text{‘offer’ never in ham} \ \bigstar$$

Prediction 1 (MLE)

Consider the email

$$x^{(A)} \;=\; (\texttt{free} = 1,\; \texttt{click} = 0,\; \texttt{meeting} = 0,\; \texttt{report} = 0,\; \texttt{offer} = 1).$$

Recall that $P(x_j = 0 \mid y = k) = 1 - \hat{\phi}_{j|k}$.

Spam score

$$\begin{aligned}
&P(x^{(A)} \mid y = 1) \cdot P(y = 1) \\
&= \hat{\phi}_{\texttt{free}|1} \cdot \big(1 - \hat{\phi}_{\texttt{click}|1}\big) \cdot \big(1 - \hat{\phi}_{\texttt{meeting}|1}\big) \cdot \big(1 - \hat{\phi}_{\texttt{report}|1}\big) \cdot \hat{\phi}_{\texttt{offer}|1} \cdot \hat{\phi}_y \\
&= \tfrac{2}{3} \cdot \tfrac{1}{3} \cdot (1 - 0) \cdot (1 - 0) \cdot \tfrac{2}{3} \cdot 0.5 \;=\; \tfrac{2}{3} \cdot \tfrac{1}{3} \cdot 1 \cdot 1 \cdot \tfrac{2}{3} \cdot 0.5 \;=\; \tfrac{4}{9} \times 0.5 \;=\; \mathbf{0.222}
\end{aligned}$$

Ham score

$$\begin{aligned}
&P(x^{(A)} \mid y = 0) \cdot P(y = 0) \\
&= \hat{\phi}_{\texttt{free}|0} \cdot \big(1 - \hat{\phi}_{\texttt{click}|0}\big) \cdot \big(1 - \hat{\phi}_{\texttt{meeting}|0}\big) \cdot \big(1 - \hat{\phi}_{\texttt{report}|0}\big) \cdot \hat{\phi}_{\texttt{offer}|0} \cdot (1 - \hat{\phi}_y) \\
&= 0 \cdot 1 \cdot \tfrac{1}{3} \cdot \tfrac{1}{3} \cdot 0 \cdot 0.5 \;=\; \mathbf{0}
\end{aligned}$$

Prediction for $x^{(A)}$ (MLE)

Since $0.222 > 0$, we predict $\hat{y} = 1$ **(spam).** ✓

Prediction 2 (MLE): A Case That Breaks

Now consider an email that mixes words from both classes – entirely realistic in practice. A work email could advertise a "free" lunch at an upcoming "meeting", or a phishing email could mimic a legitimate report.

$$x^{(B)} \;=\; (\texttt{free} = 1,\; \texttt{click} = 0,\; \texttt{meeting} = 1,\; \texttt{report} = 0,\; \texttt{offer} = 1).$$

This email contains the spam indicators `free` and `offer`, and also the ham indicator `meeting`.

Spam score

$$\begin{aligned}
&P(x^{(B)} \mid y = 1) \cdot P(y = 1) \\
&= \hat{\phi}_{\texttt{free}|1} \cdot \left(1 - \hat{\phi}_{\texttt{click}|1}\right) \cdot \hat{\phi}_{\texttt{meeting}|1} \cdot \left(1 - \hat{\phi}_{\texttt{report}|1}\right) \cdot \hat{\phi}_{\texttt{offer}|1} \cdot \hat{\phi}_y \\
&= \tfrac{2}{3} \cdot \tfrac{1}{3} \cdot \underbrace{0}_{\hat{\phi}_{\texttt{meeting}|1}=0} \cdot 1 \cdot \tfrac{2}{3} \cdot 0.5 \;=\; \mathbf{0}
\end{aligned}$$

Ham score

$$\begin{aligned}
&P(x^{(B)} \mid y = 0) \cdot P(y = 0) \\
&= \hat{\phi}_{\texttt{free}|0} \cdot \left(1 - \hat{\phi}_{\texttt{click}|0}\right) \cdot \hat{\phi}_{\texttt{meeting}|0} \cdot \left(1 - \hat{\phi}_{\texttt{report}|0}\right) \cdot \hat{\phi}_{\texttt{offer}|0} \cdot \left(1 - \hat{\phi}_y\right) \\
&= \underbrace{0}_{\hat{\phi}_{\texttt{free}|0}=0} \cdot 1 \cdot \tfrac{2}{3} \cdot \tfrac{1}{3} \cdot \underbrace{0}_{\hat{\phi}_{\texttt{offer}|0}=0} \cdot 0.5 \;=\; \mathbf{0}
\end{aligned}$$

Prediction for $x^{(B)}$ (MLE) – Classifier breaks down

Both scores are exactly zero:

$$P(x^{(B)} \mid y = 1) \cdot P(y = 1) \;=\; 0 \qquad \text{and} \qquad P(x^{(B)} \mid y = 0) \cdot P(y = 0) \;=\; 0.$$

The scores are tied at zero and no prediction can be made.

The spam score is killed by $\hat{\phi}_{\texttt{meeting}|1} = 0$: the model assigns zero probability to any spam email containing "meeting", so the entire product collapses, regardless of the strong spam evidence from `free` and `offer`. Simultaneously, the ham score is killed by $\hat{\phi}_{\texttt{free}|0} = \hat{\phi}_{\texttt{offer}|0} = 0$. A single unseen word zeroes out the entire likelihood – this is the **zero-probability problem**.

The root cause is **data sparsity**. With only 3 spam emails and a 5-word vocabulary, many word–class combinations never appear in training. The MLE faithfully reports this as probability zero – but zero is catastrophic inside a product, because it destroys all other evidence regardless of magnitude.

This is not a modelling flaw; it is an *estimation* flaw. The true probability that a spam email contains "meeting" is almost certainly not zero – we just have not seen it yet. With a real vocabulary of 50,000 words and a few thousand training emails, the vast majority of word–class combinations will be unseen, making this a severe and pervasive problem.

The fix: Laplace smoothing

We *pretend we saw every word at least once in every class* by adding a pseudo-count of $\alpha = 1$ to each cell:

$$\hat{\phi}^{(\text{smooth})}_{j|k} = \frac{\sum_{i=1}^{N} \mathbf{1}[x_j^{(i)} = 1,\ y^{(i)} = k] + 1}{\sum_{i=1}^{N} \mathbf{1}[y^{(i)} = k] + 2}.$$

The $+1$ in the numerator ensures no probability is ever zero. The $+2$ in the denominator – one for each possible value $x_j \in \{0, 1\}$ – ensures that $P(x_j = 1 \mid y = k)$ and $P(x_j = 0 \mid y = k)$ still sum to 1. The prior $\hat{\phi}_y$ is unchanged.

Laplace-Smoothed Estimates

Prior (unchanged)

$$\hat{\phi}_y = \frac{3}{6} = 0.5$$

Smoothed spam likelihoods $(y = 1)$ – denominator: $3 + 2 = 5$

$$\begin{aligned}
\hat{\phi}_{\texttt{free}|1} &= \tfrac{2+1}{3+2} = \tfrac{3}{5} = 0.600 \\
\hat{\phi}_{\texttt{click}|1} &= \tfrac{2+1}{3+2} = \tfrac{3}{5} = 0.600 \\
\hat{\phi}_{\texttt{meeting}|1} &= \tfrac{0+1}{3+2} = \tfrac{1}{5} = 0.200 \quad (\text{was } 0.000\ \checkmark) \\
\hat{\phi}_{\texttt{report}|1} &= \tfrac{0+1}{3+2} = \tfrac{1}{5} = 0.200 \quad (\text{was } 0.000\ \checkmark) \\
\hat{\phi}_{\texttt{offer}|1} &= \tfrac{2+1}{3+2} = \tfrac{3}{5} = 0.600
\end{aligned}$$

Smoothed ham likelihoods $(y = 0)$ – denominator: $3 + 2 = 5$

$$\begin{aligned}
\hat{\phi}_{\texttt{free}|0} &= \tfrac{0+1}{3+2} = \tfrac{1}{5} = 0.200 \quad (\text{was } 0.000\ \checkmark) \\
\hat{\phi}_{\texttt{click}|0} &= \tfrac{0+1}{3+2} = \tfrac{1}{5} = 0.200 \quad (\text{was } 0.000\ \checkmark) \\
\hat{\phi}_{\texttt{meeting}|0} &= \tfrac{2+1}{3+2} = \tfrac{3}{5} = 0.600 \\
\hat{\phi}_{\texttt{report}|0} &= \tfrac{2+1}{3+2} = \tfrac{3}{5} = 0.600 \\
\hat{\phi}_{\texttt{offer}|0} &= \tfrac{0+1}{3+2} = \tfrac{1}{5} = 0.200 \quad (\text{was } 0.000\ \checkmark)
\end{aligned}$$

All five previously zero parameters now carry a small but positive value.

Predictions Revisited with Laplace Smoothing

Email $x^{(A)}$: `free`=1, `click`=0, `meeting`=0, `report`=0, `offer`=1

Spam score:

$$\begin{aligned}&P(x^{(A)} \mid y=1)\cdot P(y=1)\\ &= 0.6\cdot(1-0.6)\cdot(1-0.2)\cdot(1-0.2)\cdot 0.6\cdot 0.5\\ &= 0.6\times 0.4\times 0.8\times 0.8\times 0.6\times 0.5 \;=\; \mathbf{0.04608}\end{aligned}$$

Ham score:

$$\begin{aligned}&P(x^{(A)} \mid y=0)\cdot P(y=0)\\ &= 0.2\cdot(1-0.2)\cdot(1-0.6)\cdot(1-0.6)\cdot 0.2\cdot 0.5\\ &= 0.2\times 0.8\times 0.4\times 0.4\times 0.2\times 0.5 \;=\; \mathbf{0.00256}\end{aligned}$$

Prediction for $x^{(A)}$ (Laplace)

$0.04608 > 0.00256$, so we predict $\hat{y}=1$ **(spam).** ✓ Same outcome as MLE, but now both scores are strictly positive.

Email $x^{(B)}$: `free`=1, `click`=0, `meeting`=1, `report`=0, `offer`=1

This was the email that broke the MLE classifier entirely.

Spam score:

$$\begin{aligned}&P(x^{(B)} \mid y=1)\cdot P(y=1)\\ &= \hat{\phi}_{\texttt{free}|1}\cdot(1-\hat{\phi}_{\texttt{click}|1})\cdot\hat{\phi}_{\texttt{meeting}|1}\cdot(1-\hat{\phi}_{\texttt{report}|1})\cdot\hat{\phi}_{\texttt{offer}|1}\cdot\hat{\phi}_y\\ &= 0.6\times 0.4\times 0.2\times 0.8\times 0.6\times 0.5 \;=\; \mathbf{0.01152}\end{aligned}$$

Ham score:

$$\begin{aligned}&P(x^{(B)} \mid y=0)\cdot P(y=0)\\ &= \hat{\phi}_{\texttt{free}|0}\cdot(1-\hat{\phi}_{\texttt{click}|0})\cdot\hat{\phi}_{\texttt{meeting}|0}\cdot(1-\hat{\phi}_{\texttt{report}|0})\cdot\hat{\phi}_{\texttt{offer}|0}\cdot(1-\hat{\phi}_y)\\ &= 0.2\times 0.8\times 0.6\times 0.4\times 0.2\times 0.5 \;=\; \mathbf{0.00384}\end{aligned}$$

Prediction for $x^{(B)}$ (Laplace)

$0.01152 > 0.00384$, so we predict $\hat{y}=1$ **(spam).** ✓
Both scores are strictly positive and the classifier can now make a decision. The presence of `meeting` penalises the spam score through the small factor $\hat{\phi}_{\texttt{meeting}|1} = 0.2$, but the combined evidence of `free` and `offer` still tips the balance toward spam. Compare to MLE, where both scores were identically zero and no prediction was possible.

Side-by-Side Parameter Comparison

Parameter	MLE formula	MLE value	Laplace formula	Laplace value
$\hat{\phi}_y$	3/6	0.500	3/6	0.500
$\hat{\phi}_{\texttt{free}\|1}$	2/3	0.667	(2+1)/(3+2)	0.600
$\hat{\phi}_{\texttt{click}\|1}$	2/3	0.667	(2+1)/(3+2)	0.600
$\hat{\phi}_{\texttt{meeting}\|1}$	0/3	**0.000** ★	(0+1)/(3+2)	0.200
$\hat{\phi}_{\texttt{report}\|1}$	0/3	**0.000** ★	(0+1)/(3+2)	0.200
$\hat{\phi}_{\texttt{offer}\|1}$	2/3	0.667	(2+1)/(3+2)	0.600
$\hat{\phi}_{\texttt{free}\|0}$	0/3	**0.000** ★	(0+1)/(3+2)	0.200
$\hat{\phi}_{\texttt{click}\|0}$	0/3	**0.000** ★	(0+1)/(3+2)	0.200
$\hat{\phi}_{\texttt{meeting}\|0}$	2/3	0.667	(2+1)/(3+2)	0.600
$\hat{\phi}_{\texttt{report}\|0}$	2/3	0.667	(2+1)/(3+2)	0.600
$\hat{\phi}_{\texttt{offer}\|0}$	0/3	**0.000** ★	(0+1)/(3+2)	0.200

Key observations

1. $\hat{\phi}_y$ is unchanged by smoothing (no zero-probability issue in the prior).
2. Non-zero MLE estimates shrink slightly toward 0.5, e.g. $\frac{2}{3} \approx 0.667 \rightarrow \frac{3}{5} = 0.600$, as probability mass is redistributed to unseen events.
3. The five zero MLE entries are raised to 0.200, eliminating the zero-product problem entirely.
4. With only 3 training emails per class, smoothing has a pronounced effect. With more data the correction becomes negligible: as $N \to \infty$, $\hat{\phi}_{j|k}^{(\text{smooth})} \to \hat{\phi}_{j|k}^{(\text{MLE})}$.